

Setup & protocol

- **Backbones (frozen):** DINOv3-S/B/L, MAE-L, SigLIP2-L, V-JEPA2.1-L, PESpatial-L; all ViT blocks used.
- **Benchmarks:** NYUv2 surface normals, depth (log-depth head), semantic segmentation; Taskonomy (aggregate L_1); NAVI camera pose.
- **Baselines:** LastLayer; AvgLayer; WeightedLayers (one learned layer mix); SpatialWeights (learned mix per patch position, input-independent); ConcatLayers (concat $\rightarrow 1\times 1$ projection + GELU).
- **AMDF:** per patch, 16 experts with per-layer queries; keys $W_k(h_\ell + p_\ell)$ in 64-d, softmax over layers ($\sqrt{64}$); values $W_v h_\ell$ at full width, split in 16 blocks; no bottleneck, no per-layer refinement, no MLP in the probe.
- **Heads:** dense tasks — linear 1×1 conv then bilinear upsampling; pose — pool both images, concatenate, MLP. Identical for all methods within a table.
- **Protocol:** same recipe for every row of a table; mean of up to 3 seeds (8/42/123; a few baseline cells have 1–2); best-epoch validation metric.
- **Ablation (per-layer vs shared queries):** paired gains, same seeds, mean over 7 backbones — normals +1.62 d_1 (7/7), Taskonomy $-0.73 L_1\times 100$ (7/7), seg +0.16 mIoU (6/7), depth +0.51 d_1 (5/7), pose -2.37° rot. (5/7; mostly SigLIP2-L, -13°); cost +23.5k params ($\approx 2\%$).

NYUv2 surface normals — d_1 (%) $\uparrow$

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	22.31	24.52	23.50	32.81	13.22	36.42	16.88	1.05	5.43
AvgLayer	20.74	27.89	30.99	35.32	15.17	32.63	17.42	1.05	5.29
WeightedLayers	25.44	30.80	34.65	36.58	16.74	35.52	17.46	1.05	4.14
SpatialWeights	25.50	31.26	34.84	37.53	17.36	36.14	19.95	1.06	3.14
ConcatLayers	40.20	45.19	48.48	49.87	36.12	49.10	33.41	26.22	1.00
AMDF (ours)	37.31	41.98	45.55	46.90	32.61	46.34	29.58	1.17	2.00

NYUv2 depth — d_1 (%) $\uparrow$ (log-depth head)

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	58.80	61.73	63.17	58.10	55.16	67.63	54.11	1.05	4.14
AvgLayer	51.90	60.07	60.75	59.30	57.51	64.46	54.93	1.05	5.57
WeightedLayers	56.13	61.34	62.15	59.77	57.55	66.07	55.13	1.05	4.00
SpatialWeights	55.52	61.44	62.11	59.67	58.41	64.93	55.67	1.05	4.14
ConcatLayers	61.13	64.86	67.35	64.11	62.28	67.44	57.62	26.22	1.71
AMDF (ours)	58.81	64.06	66.47	64.22	63.01	69.80	58.56	1.16	1.43

NYUv2 semantic segmentation — mIoU (%) $\uparrow$

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	44.20	53.41	57.08	38.94	40.39	44.05	32.79	1.09	3.86
AvgLayer	35.49	47.48	50.80	40.54	42.13	41.94	33.36	1.09	5.14
WeightedLayers	40.26	50.02	52.78	41.00	42.11	42.94	33.25	1.09	4.14
SpatialWeights	39.79	49.72	52.50	40.94	42.19	42.69	33.34	1.09	4.43
ConcatLayers	44.61	52.63	56.50	43.89	47.07	46.26	36.48	26.26	1.57
AMDF (ours)	44.34	53.13	57.17	42.36	43.83	44.77	33.64	1.20	1.86

Taskonomy — aggregate $L_1 \times 100 \downarrow$

Method	DINOv3-S	DINOv3-B	DINOv3-L	MAE-L	SigLIP2-L	V-JEPA2.1-L	PESpatial-L	Params (M)	Rank
LastLayer	41.64	39.21	39.26	39.50	46.81	36.94	42.21	1.06	5.29
AvgLayer	43.66	39.45	38.51	38.60	45.48	37.84	41.83	1.06	5.43
WeightedLayers	41.47	38.51	37.51	38.19	44.30	37.32	41.70	1.06	3.29
SpatialWeights	41.60	38.61	37.64	38.25	44.36	37.34	41.52	1.06	4.00
ConcatLayers	33.90	32.19	30.95	32.67	35.94	31.40	34.79	26.23	1.00
AMDF (ours)	37.49	33.86	32.21	33.77	37.05	34.78	36.83	1.17	2.00

NAVI camera pose — rotation ($^\circ$) $\downarrow$ / translation $\downarrow$

Method	DINOv3-S		DINOv3-B		DINOv3-L		MAE-L		SigLIP2-L		V-JEPA2.1-L		PESpatial-L	
	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.	Rot.	Trans.
LastLayer	45.86	0.17	44.24	0.17	40.63	0.16	38.61	0.16	62.62	0.22	37.34	0.15	65.16	0.23
AvgLayer	44.76	0.17	36.04	0.15	36.44	0.14	37.28	0.15	59.01	0.21	47.82	0.19	66.44	0.24
WeightedLayers	43.71	0.17	35.25	0.15	35.09	0.14	35.82	0.15	58.60	0.21	43.59	0.17	66.21	0.23
SpatialWeights	44.49	0.17	35.66	0.15	35.94	0.14	35.49	0.15	54.83	0.20	44.21	0.17	66.29	0.23
ConcatLayers	34.01	0.14	26.55	0.12	24.31	0.11	31.10	0.13	32.29	0.14	37.46	0.15	52.28	0.19
AMDF (ours)	40.90	0.16	31.61	0.14	31.80	0.13	31.70	0.14	47.37	0.18	36.72	0.15	63.77	0.22

Build red = best, underlined = second best per column (among the 6 rows). Rank = mean over backbone columns of the per-column rank (ties averaged; NAVI over 14 rot/trans sub-columns). Params (M) = trainable parameters (probe + task head), DINOv3-L, SpatialWeights = learned layer mix per patch position (input-independent); ConcatLayers = concat $\rightarrow 1\times 1$ projection + GELU. Full-width setup: no channel reduction, 64-d depth keys/queries, per-layer expert queries (AMDF), 16 experts; frozen backbones; same recipe for every row of a table; mean of up to 3 seeds (8/42/123; a few baseline cells have 1–2), best-epoch validation metric. Depth uses a log-depth head.